Requirement Specification

REQ-0224 — Performance — Battery Optimization

1. Identification

Requirement ID	REQ-0224
Title	Performance — Battery Optimization
Type	Reliability
Priority	Medium
Severity	Critical
Status	Approved
Version	v3.0
Created	2022-02-01
Updated	2024-02-19
Source Document	SRS-APP-011-v5.pdf

2. Application & Solution

Application ID	APP-011
Application	ConnectedDevices
Domain	IoT
App Version	4.4.4
Solution ID	SOL-020
Solution	Playlist Manager
Component	Network Stack
Feature	voice command recognition

3. Requirement Description

The system shall validate crash reporting without data loss under stress-test conditions.

4. Acceptance Criteria

AC-1: The system shall satisfy the requirement within the defined performance SLA (p95 < 500ms).

AC-2: All failure scenarios must return a human-readable error message to the user.

AC-3: All events must be logged to the audit trail with a valid timestamp and user context.

AC-4: The feature must operate correctly in all supported environments (Mobile, Wearable, Cloud).

AC-5: Security constraints: all data in transit must be encrypted using TLS 1.3 or higher.

5. Dependencies

This requirement depends on the Playlist Manager (SOL-020) being operational and at version 2.5 or above.

SYNTHETIC DATA — No real Samsung confidential information

6. Related Test Cases

TC-1492	Confirm that background sync triggers the expected notificat...	Negative	High	Active
TC-1767	Verify that playlist shuffle completes successfully when the...	Smoke	Critical	Draft
TC-1810	Ensure that voice command recognition does not exceed the me...	Security	Medium	Active

8. Synthetic Data Notice

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